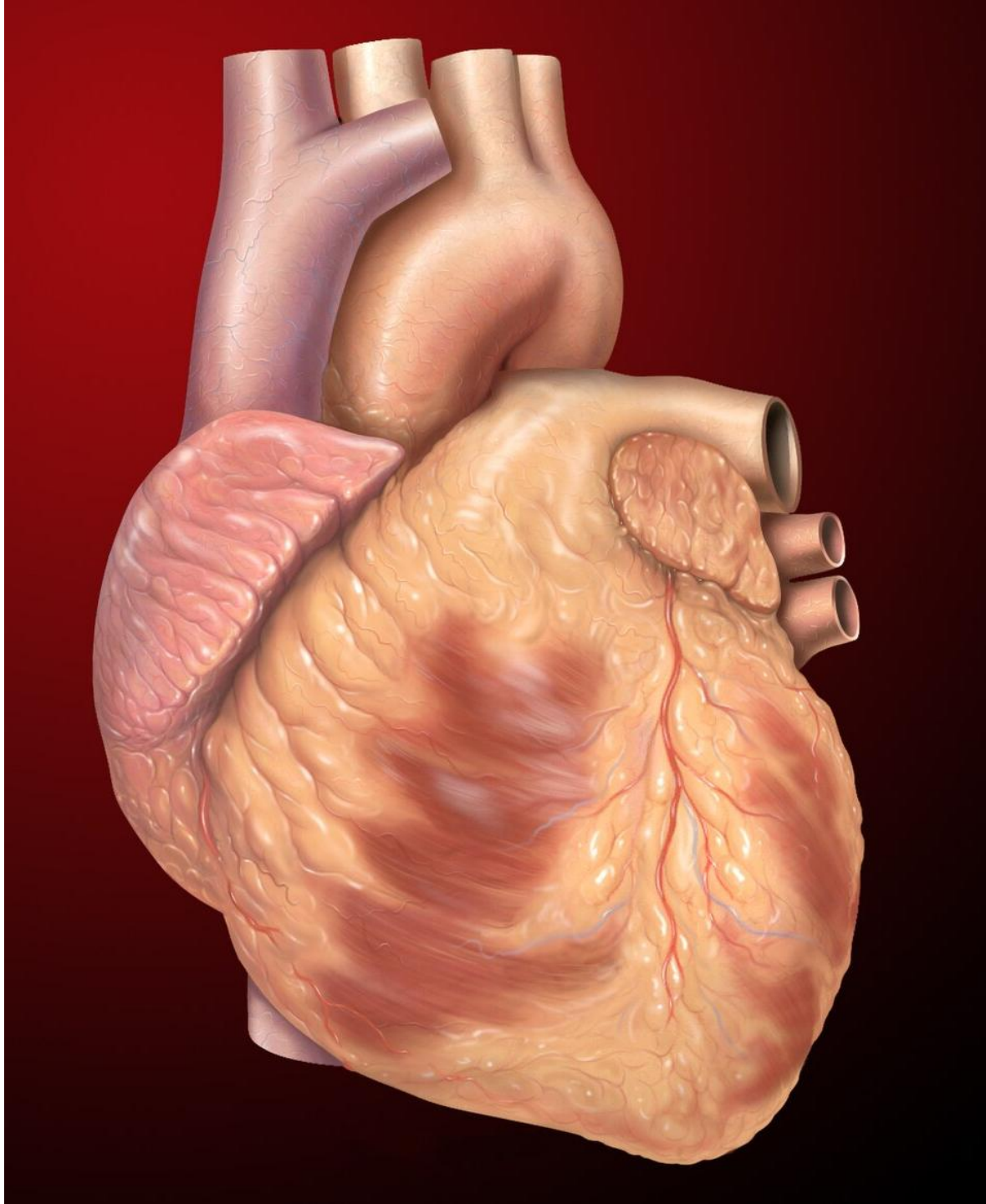
DP Biology – B3.2

Transport

- Which adaptations help animals and plants move fluids around?
- How does transport in animals compare with transport in plants?

Animal statements B3.2.1–6, SL and HL. Textbook pp. 295–302. Plants are in Plant Systems.
Image: human heart, anterior view. Patrick J. Lynch, CC BY 2.5.

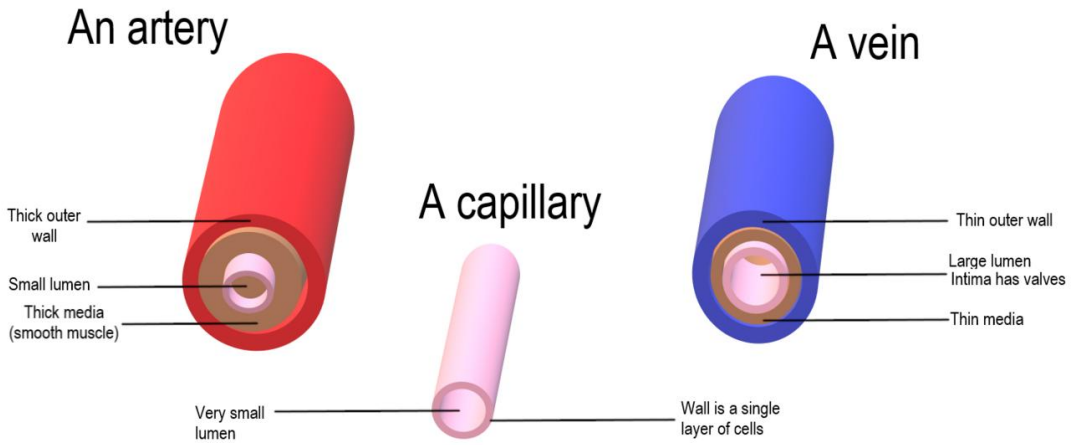


Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.2.1–2	Explain how capillaries suit exchange, and identify arteries and veins in micrographs
2	B3.2.3–5	Explain how arteries and veins are adapted to their pressures, and measure a pulse rate
3	B3.2.6	Explain the causes and consequences of coronary occlusion, and evaluate data on heart disease

Three types of blood vessel

B3.2.1–2



Artery, capillary and vein compared. Christinelmiller, CC BY-SA 4.0.

Artery away from the heart

up to 120 mm Hg

Capillary exchange with tissues

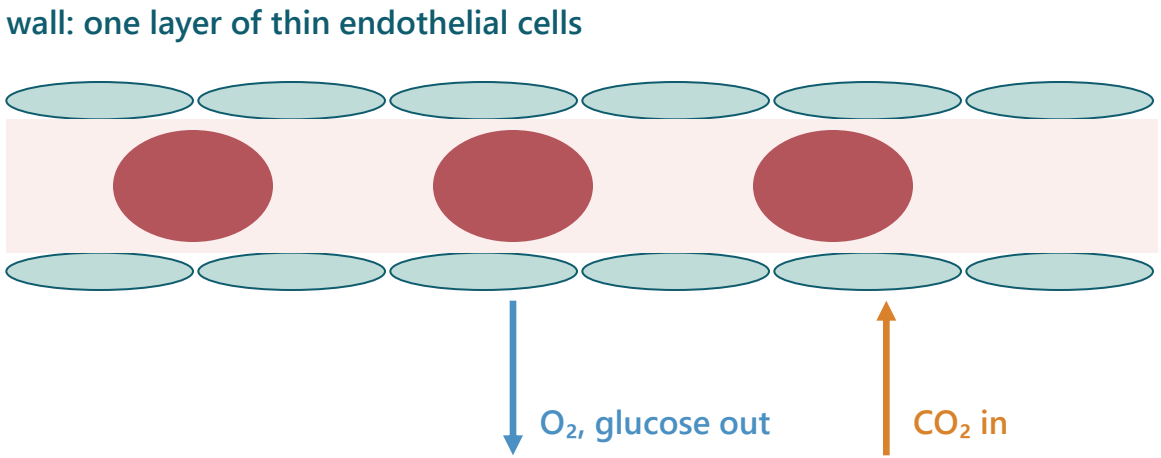
about 35 mm Hg

Vein back to the heart

about 5 mm Hg

Blood flows artery → capillary → vein, and pressure falls all the way.

Capillaries are built for exchange



Drawn, not to scale. Red cells only just fit the lumen (about 8 µm).

Wall one cell thick: short diffusion distance.

Narrow lumen: red cells pass close to the wall.

Branching network: huge surface area, and no cell is far from a capillary.

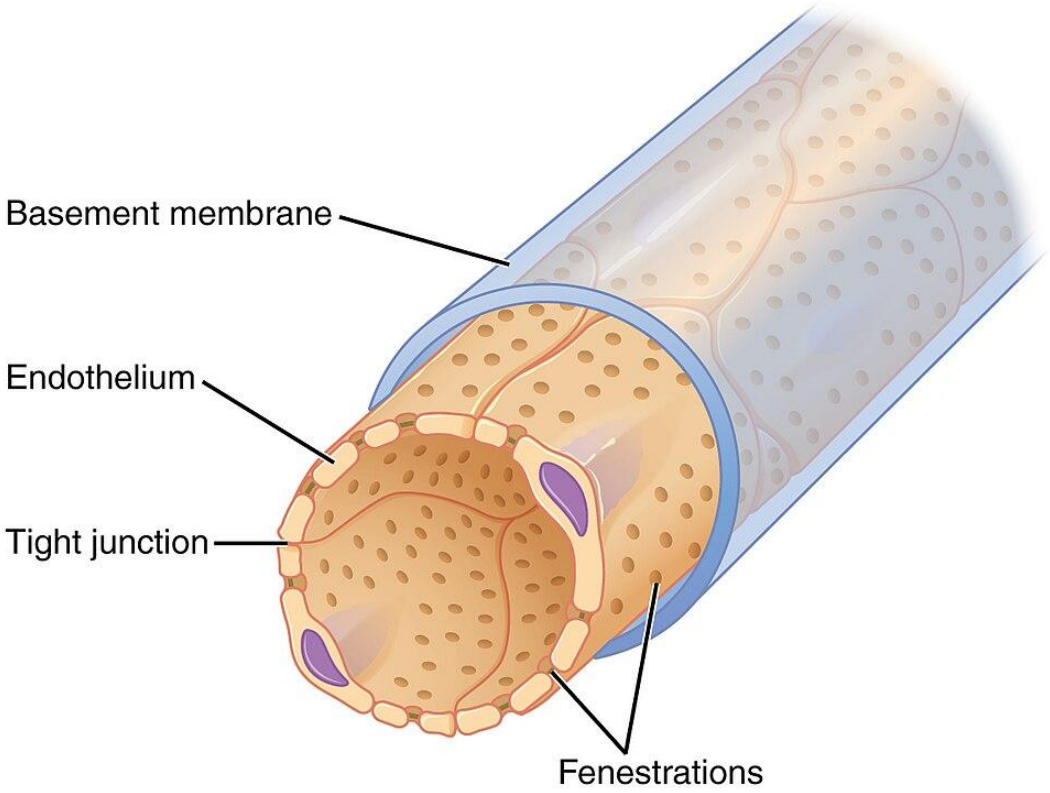
Slow flow: more time for exchange.

Key idea

Arteries and veins only carry blood. Exchange with the tissues happens in capillaries.

Fenestrated capillaries: faster exchange

B3.2.1



A fenestrated capillary. OpenStax College, CC BY 3.0.

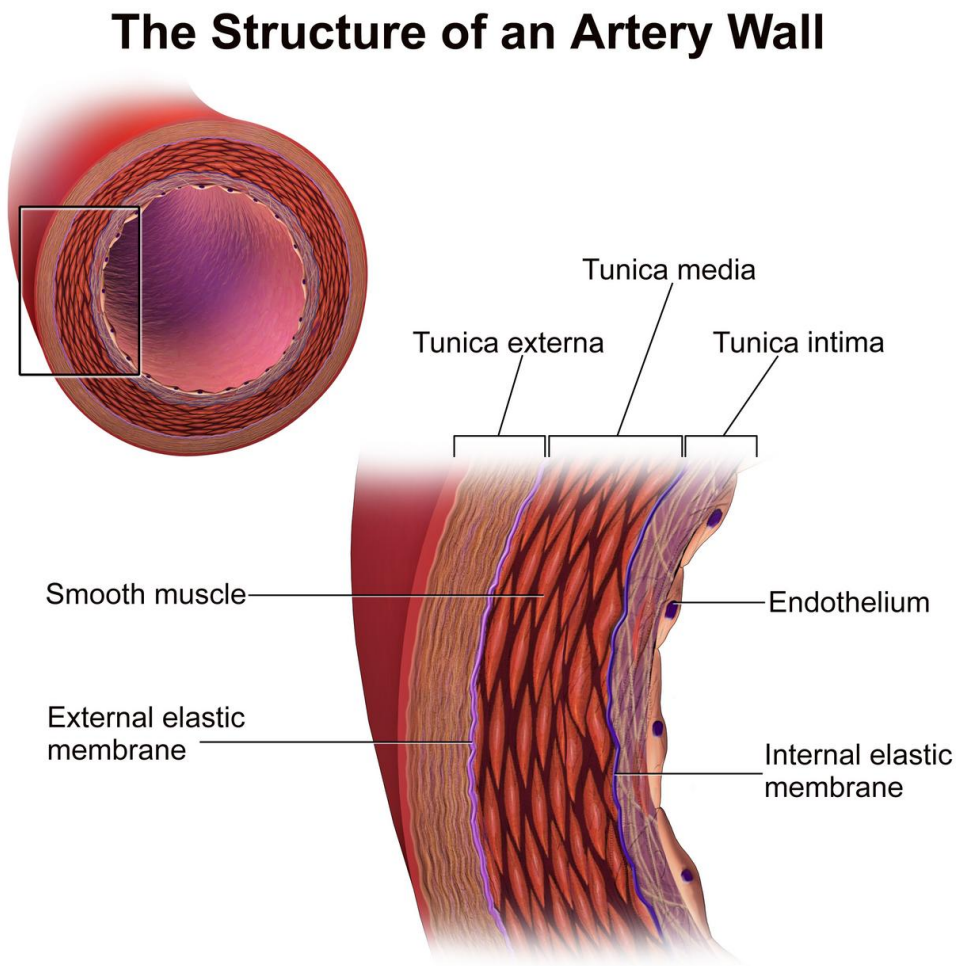
Fenestrations are small pores through the capillary wall, found where exchange must be very fast, such as the kidney and small intestine.

Plasma and small solutes leak out easily and form **tissue fluid**. Cells and large proteins mostly stay in the blood.

Think

Suggest why the capillaries in the villi of the small intestine are fenestrated.

The wall has three layers



Artery wall. BruceBlaus, CC BY 3.0.

Tunica intima

Smooth endothelium lining the lumen: little friction.

Tunica media

Smooth muscle and elastic fibres.

Tunica externa

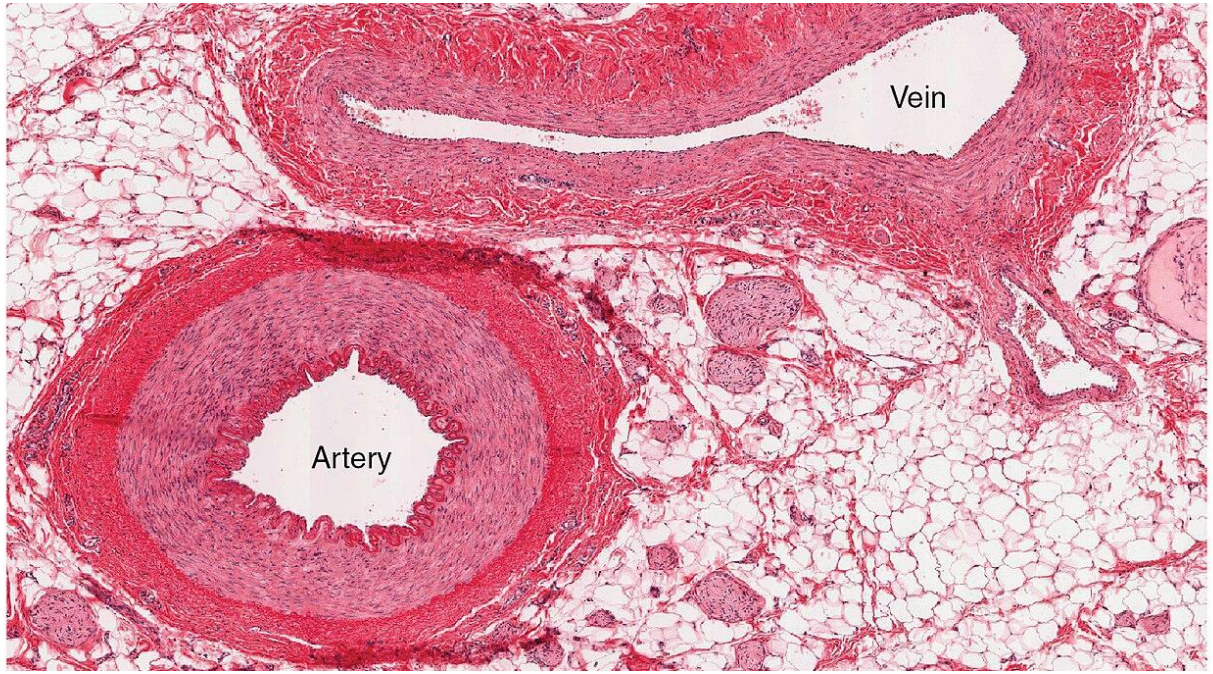
Tough connective tissue with collagen.

Key idea

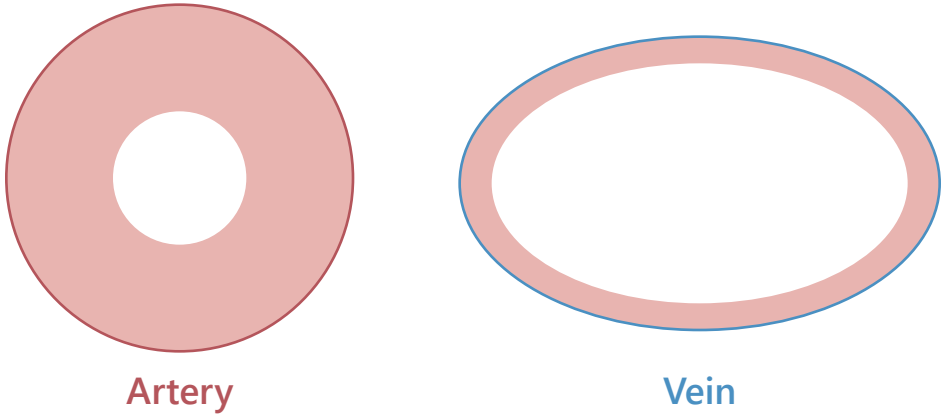
Arteries and veins have the same three layers. Their thickness differs with the pressure inside.

Identifying vessels in a micrograph

B3.2.2



Light micrograph: vein (above) and artery (below). OpenStax College, CC BY 3.0.



Compare the **wall** with the **lumen**. Artery: thick wall, narrow lumen, round. Vein: thin wall, wide lumen, often squashed.

Common mistake

Do not judge by overall size. Use the wall thickness relative to the lumen.

Arteries and veins compared

B3.2.2

	Artery	Vein
Wall relative to lumen	Thick	Thin
Lumen	Narrow	Wide
Tunica media	Thick: much muscle and elastic tissue	Thin: little muscle or elastic tissue
Valves	None (except leaving the heart)	Present, especially in the limbs
Pressure	High, in pulses	Low, no pulse

Activity

Without looking, sketch both vessels in section and label all three layers.

Practice: which vessel is it?

Measurements from sections of four vessels. Typical values.

Vessel	Wall thickness / mm	Lumen diameter / mm
P	1.0	3.0
Q	0.001	0.008
R	0.5	6.0
S	2.0	25

1. Identify P, Q, R and S as artery, capillary, vein or aorta.
2. Calculate the wall : lumen ratio for P and R.
3. Explain how the ratio helps you decide.

Check-in: class 1

B3.2.1–2

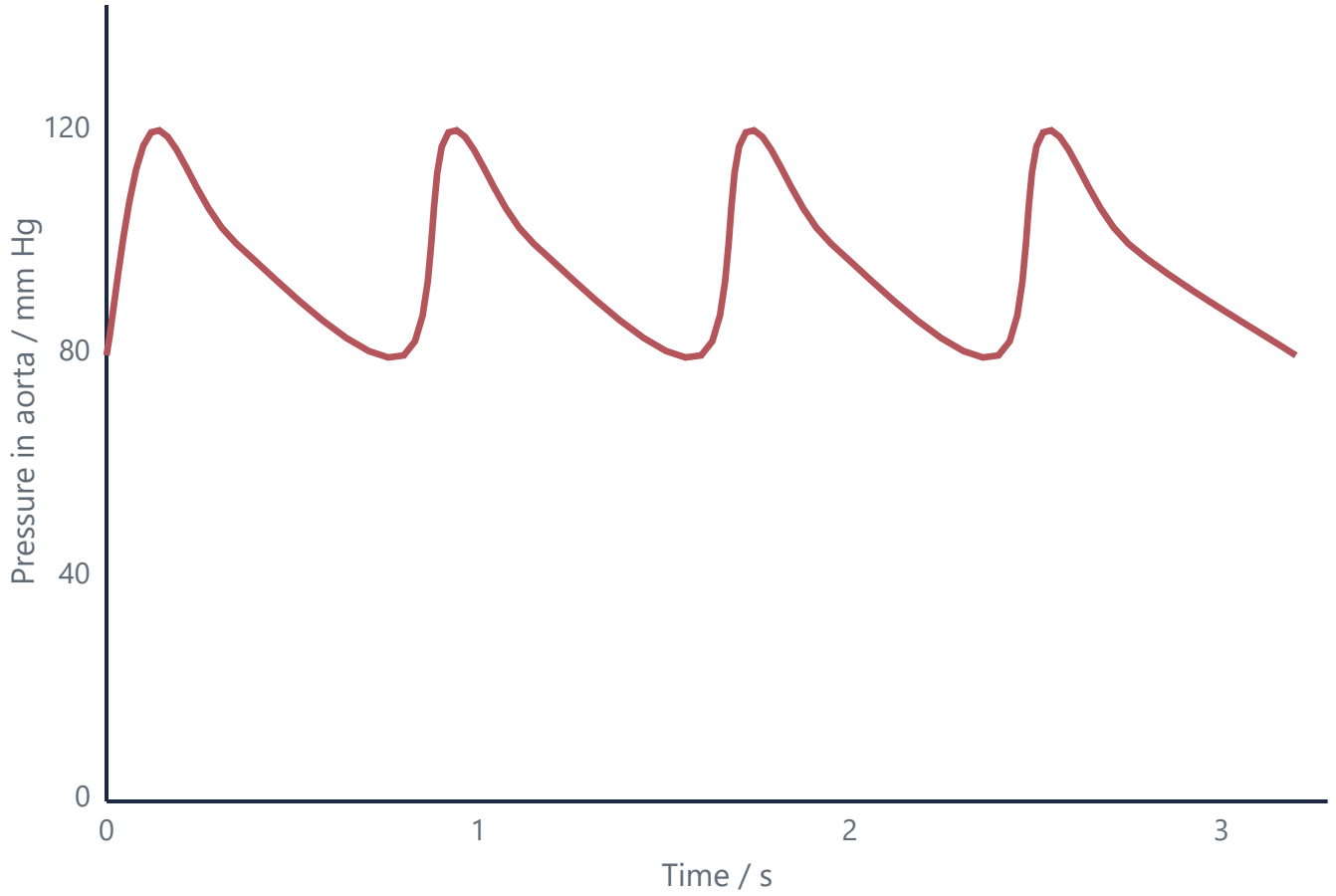
1. State three adaptations of capillaries for exchange.
2. Outline what fenestrations are and where they are found.
3. Name the three layers of an artery wall.
4. Distinguish between an artery and a vein in a micrograph.
5. Explain why exchange does not happen across artery walls.

How to use this

Answer without your notes, then check against slides 3–9.

Elastic arteries even out the pressure

B3.2.3



Modelled trace with typical values. It never drops to zero between beats.

Each beat, a surge of blood **stretches** the elastic fibres. Between beats they **recoil**, pushing blood on.

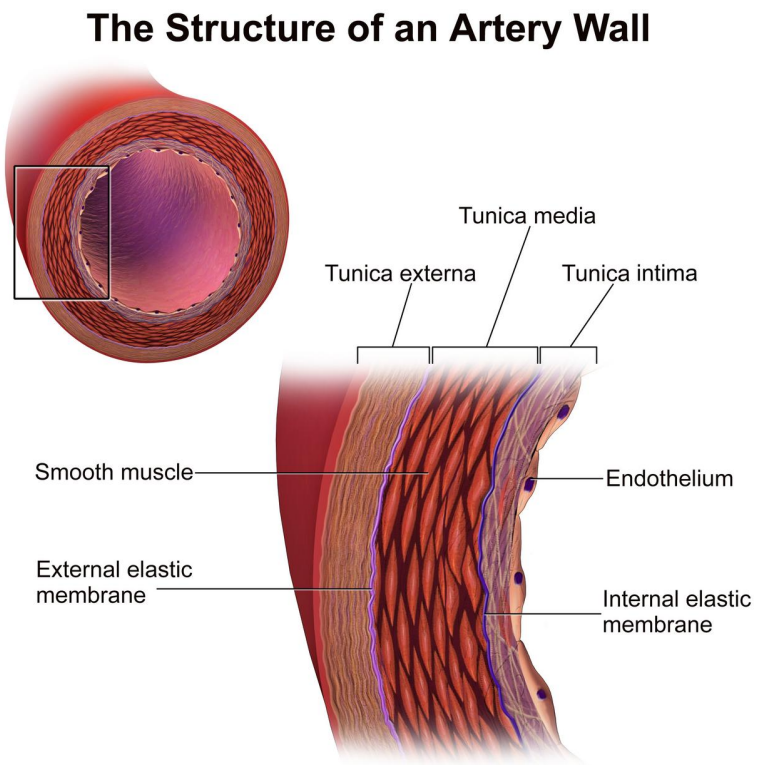
So pressure stays high and flow keeps going, from about 80 to 120 mm Hg.

Two words

Arteries **withstand** high pressure (strength) and **maintain** it (elasticity and muscle).

What each tissue in an artery does

B3.2.3



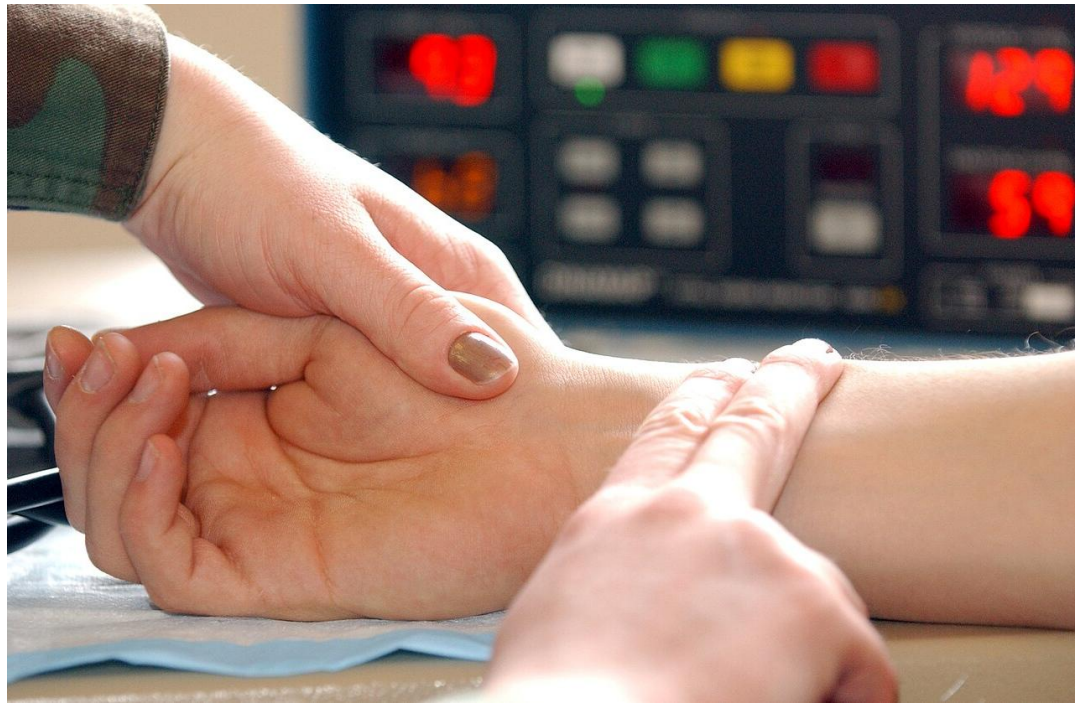
Layers of an artery wall. BruceBlaus, CC BY 3.0.

Tissue	Role in an artery
Elastic fibres	Stretch and recoil, maintaining pressure
Smooth muscle	Contracts to narrow the lumen; helps keep pressure up
Collagen	Strong: stops the wall bursting
Thick wall overall	Withstands high pressure

Link

Arterioles have the most smooth muscle. They control how much blood reaches each capillary bed.

Taking a pulse



Feeling the radial pulse. Jeanette Copeland, US Air Force, public domain.

Each beat sends a **pressure wave** along the arteries. Feel it where an artery lies near the skin over a bone.

Radial: wrist, thumb side. **Carotid**: side of the neck. Count for 60 s, or 30 s and double.

Common mistake

Use your fingertips, not your thumb. The thumb has a pulse of its own.

Traditional and digital methods

B3.2.4

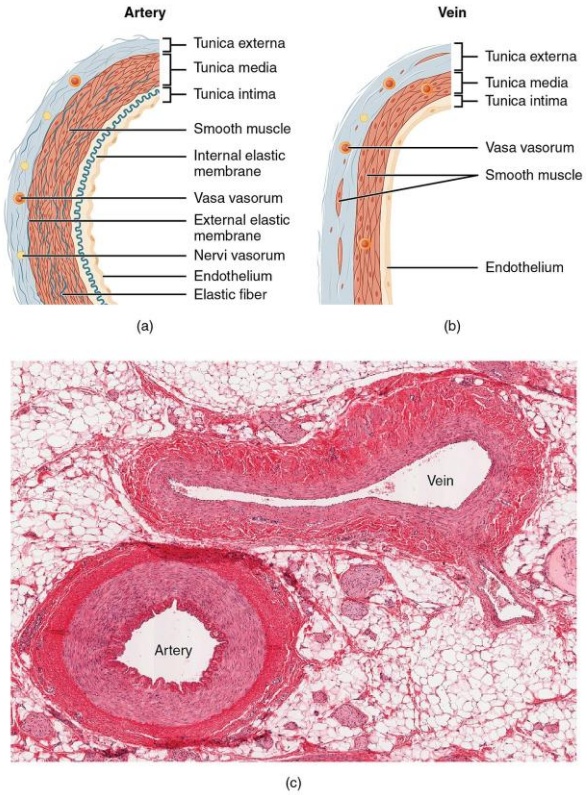
	Fingertips and a timer	Watch, oximeter or data logger
How	Feel and count the beats	Light shines into the skin; a sensor detects each pulse of blood
Advantage	Free, simple, anywhere	Continuous; records during exercise
Limitation	Counting errors; hard while moving	Cost; accuracy depends on device and fit

Activity

Measure your resting pulse both ways, three times each. Which method is more consistent? Volunteers only; anyone with a heart or breathing condition acts as timer.

Veins return blood at low pressure

B3.2.5



Artery and vein walls, with a micrograph. OpenStax College, CC BY 3.0.

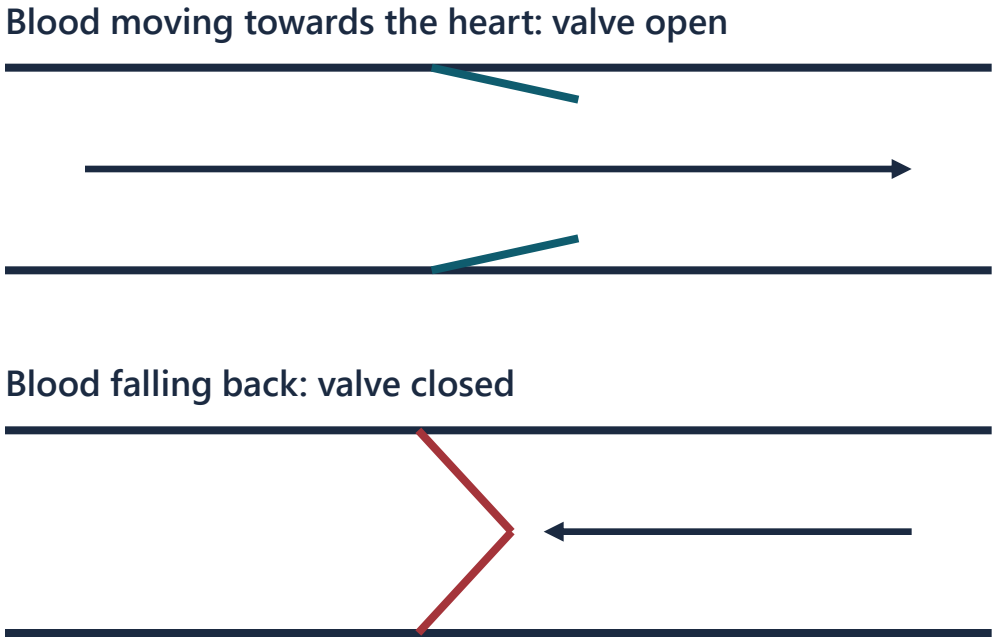
Blood loses most of its pressure in the capillaries: in veins there is **no pulse**.

A **wide lumen** cuts friction. A **thin, flexible wall** is squeezed by nearby muscles. **Valves** stop backflow.

Think

Why would a thick, rigid wall be a disadvantage for a vein?

Valves and muscle action



Drawn section of a vein, with the heart to the right.

Leg muscles contract and **squeeze** the veins, pushing blood along.

Blood moving towards the heart pushes the flaps open. Blood falling back fills the pockets and **closes** them. So flow is one way.

Link

Walking helps blood return from the legs. Standing still for hours can make feet swell or cause fainting.

Working with data: pulse and exercise

B3.2.4

One student’s pulse rate before and after two minutes of step-ups. The values are invented for practice; the pattern is realistic.

Time	Pulse / beats per min
At rest	68
1 min after	112
2 min after	96
3 min after	82
5 min after	70

1. Calculate the percentage increase in pulse rate.
2. State how long the pulse took to return close to resting.
3. Explain why the pulse rate rises during exercise.
4. Suggest why a trained athlete recovers faster.

Check-in: class 2

B3.2.3–5

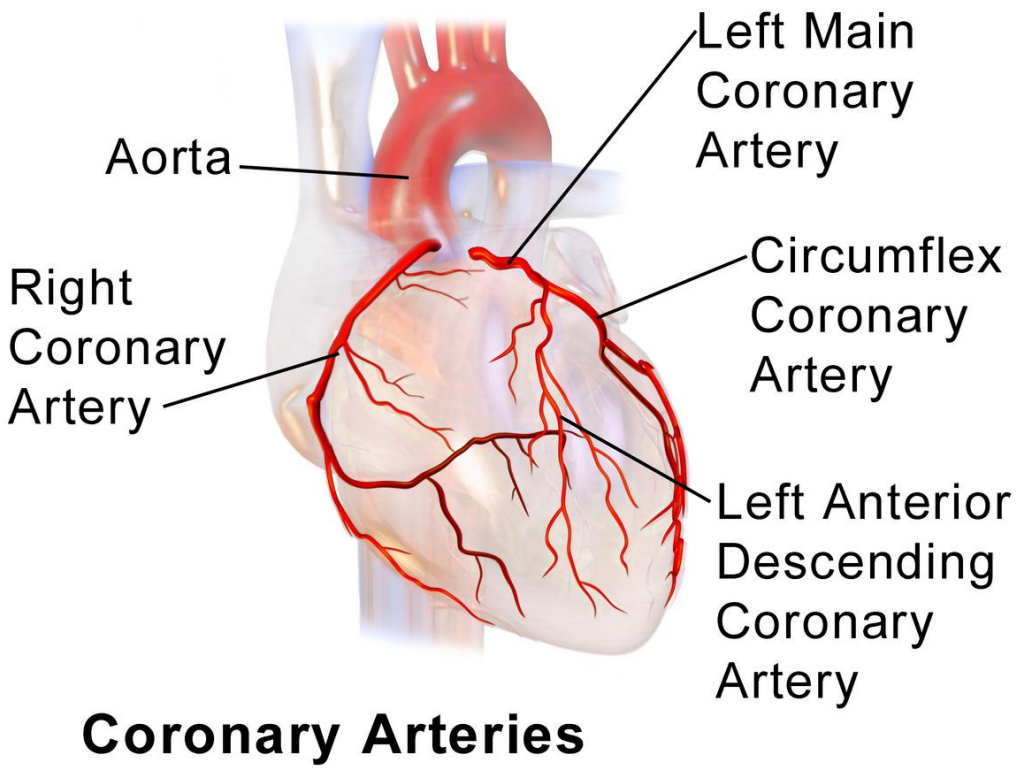
1. Explain how elastic fibres maintain pressure in arteries.
2. State the role of collagen and of smooth muscle in an artery wall.
3. Describe how to take a pulse rate by hand.
4. Explain how valves and skeletal muscle return blood to the heart.
5. Explain why veins have no pulse.

How to use this

Answer without your notes, then check against slides 11–17.

Coronary arteries supply the heart

B3.2.6



The coronary arteries. BruceBlaus, CC BY 3.0.

Heart muscle respires all the time and needs a constant supply of oxygen and glucose. The blood inside the chambers does not supply it.

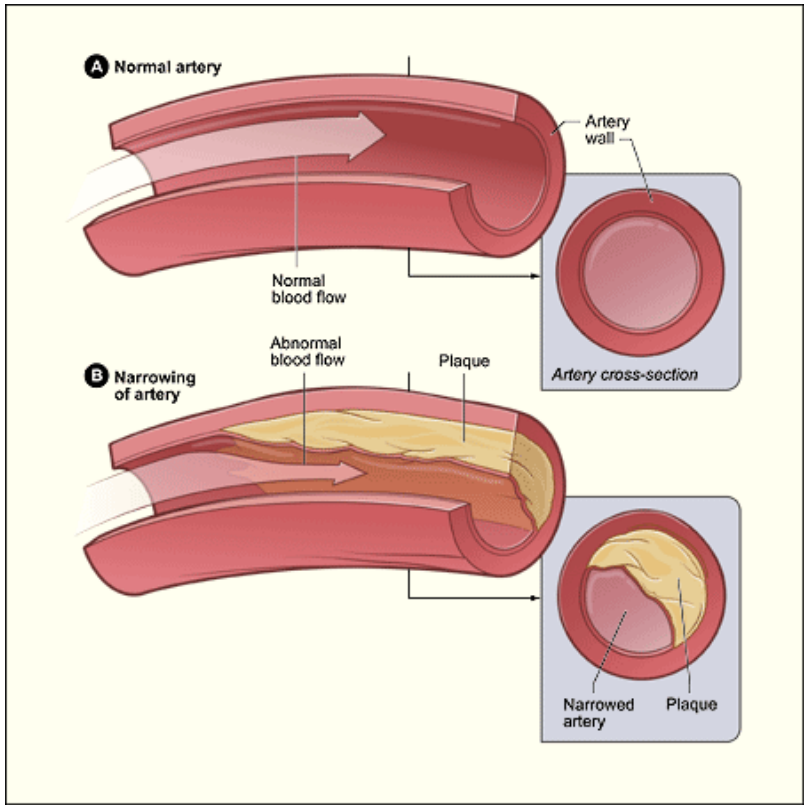
Coronary arteries branch from the aorta and spread over the heart. **Occlusion** means a blockage of a vessel.

Key idea

A block starves only the muscle downstream, but that can stop the heart pumping well.

How arteries become blocked

B3.2.6



Normal artery (A) and artery narrowed by plaque (B). NHLBI, public domain.

Fatty **atheroma** forms under the endothelium

↓

Lumen narrows; wall hardens

↓

Plaque ruptures

↓

A clot forms: a **thrombus**

↓

Artery blocked: occlusion

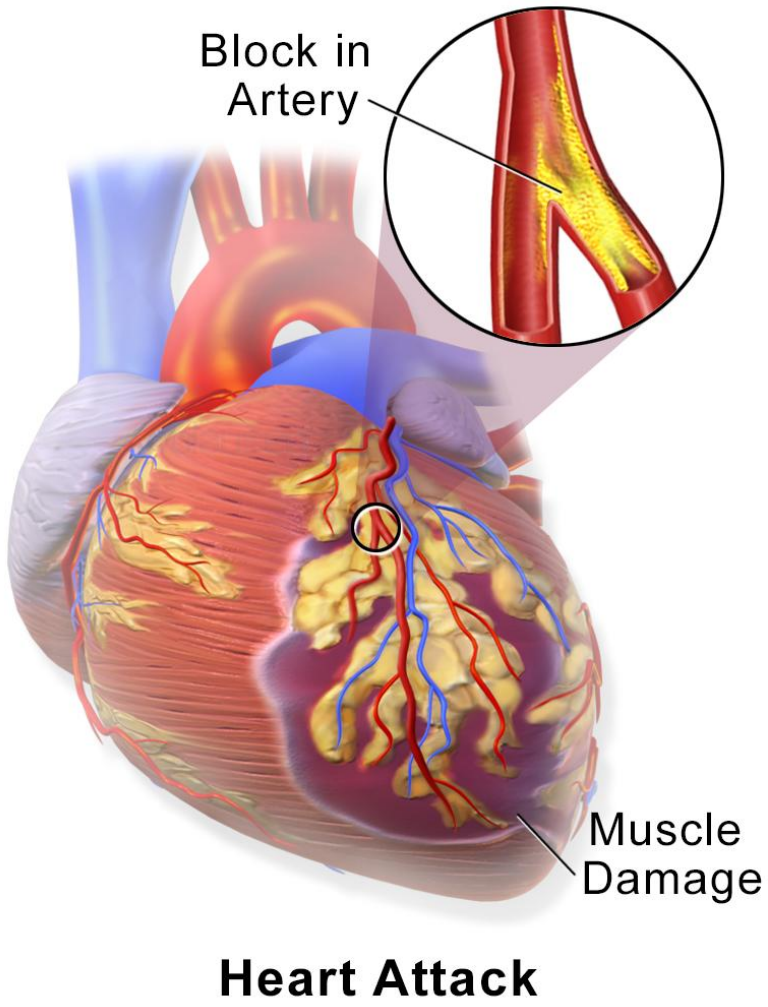
Embolus

A clot that breaks free and lodges in a smaller vessel.

Common mistake

Plaque forms within the wall, beneath the endothelium, not on top of it.

Consequences of occlusion



Blocked coronary artery and damaged muscle. Blausen Medical, CC BY 3.0.

Degree of block	What follows
Partial narrowing	Too little oxygen in exercise: chest pain (**angina**)
Complete block	Muscle downstream cannot respire aerobically, and dies
Much muscle lost	Heart cannot pump well: **heart attack** (myocardial infarction)

Treatment

Stents hold a narrowed artery open; a bypass uses a vessel from the leg to carry blood around the block.

Risk factors for coronary heart disease

B3.2.6

- High blood pressure
- Smoking
- Diet high in saturated fat
- Obesity
- Diabetes
- Lack of exercise
- Age
- Family history (genes)

Red: can be changed by lifestyle or treatment. Grey: cannot be changed.

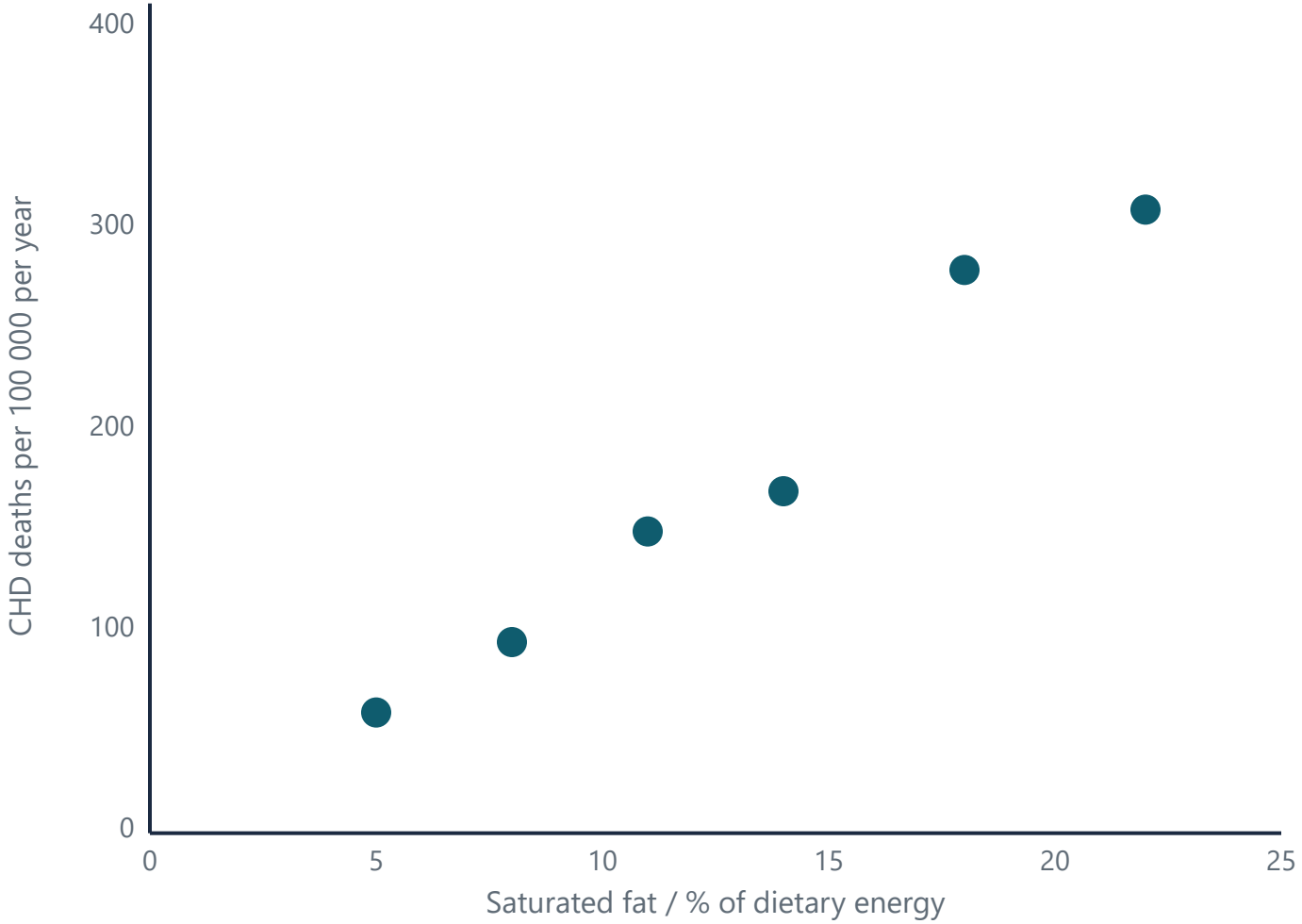
A **risk factor** raises the chance of disease. It does not mean the person will get it.

Think

Choose two risk factors and explain how each could damage coronary arteries.

Working with data: populations

B3.2.6



Invented data for six fictional countries, for practice only.

1. Describe the relationship shown.
2. Estimate the death rate at 10% saturated fat.
3. Give two other variables that could explain the pattern.
4. Explain why this does not prove that saturated fat causes CHD.

Evaluating data on heart disease

B3.2.6

Strengths to look for	Limitations to look for
Statistically significant correlation	Small or unrepresentative sample
Large sample over a long time	Other variables not controlled, such as smoking
Little spread; small error bars	Survey data depend on honest reporting
Valid measure of health	Results from animals may not apply to humans

Activity

For the graph on slide 23, write one strength and two limitations, then say what extra evidence would support a causal link.

Check-in: class 3

B3.2.6

1. Explain why heart muscle needs coronary arteries.
2. Outline how atherosclerosis leads to occlusion.
3. Distinguish between a thrombus and an embolus.
4. Explain why a blocked coronary artery can cause a heart attack.
5. Explain the difference between correlation and causation.

How to use this

Answer without your notes, then check against slides 19–24.

Common mistakes to avoid

Not this	But this
Arteries carry oxygenated blood; veins deoxygenated.	Arteries carry blood away from the heart; the pulmonary artery carries deoxygenated blood.
Capillary walls are one membrane thick.	They are one cell thick: a layer of endothelial cells.
Veins have thick muscular walls.	Veins have thin walls and a wide lumen.
The pulse is the heart beating in the wrist.	It is the stretch and recoil of an artery wall.
A heart attack is when the heart stops.	It is when heart muscle dies after a coronary artery is blocked.

Summary: blood vessels

B3.2.1–6

	Key idea
B3.2.1	Capillaries: one cell thick, narrow, branched, sometimes fenestrated
B3.2.2	Three layers; artery thick wall and narrow lumen, vein the reverse
B3.2.3	Arteries: collagen withstands, elastic fibres and muscle maintain pressure
B3.2.4	Pulse by fingertips (radial, carotid) or by digital sensor
B3.2.5	Veins: wide lumen, thin wall, valves, squeezed by muscles
B3.2.6	Atheroma, plaque, clot: occlusion starves heart muscle

Check that you can...

1. **Explain** how capillaries are adapted for exchange
2. **Identify** arteries and veins in micrographs
3. **Explain** how artery walls suit high pressure
4. **Describe** how to measure a pulse rate
5. **Explain** how veins return blood at low pressure
6. **Outline** the causes and consequences of coronary occlusion
7. **Evaluate** data on coronary heart disease

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Give one adaptation of a capillary and explain how it helps exchange.

2. Vessel X has a wall 1.2 mm thick and a lumen 4 mm wide. Vessel Y has a wall 0.4 mm thick and a lumen 5 mm wide. Which is the artery, and how can you tell?

3. In two sentences, explain how a plaque can lead to a heart attack.

Image credits

Heart anterior exterior view.jpg – Patrick J. Lynch – CC BY 2.5
Artery Vein Capillary Comparison.png – Christinelmiller – CC BY-SA 4.0
2614 Fenestrated Capillary.jpg – OpenStax College – CC BY 3.0
Blausen 0055 ArteryWallStructure.png – BruceBlaus – CC BY 3.0
2102 Comparison of Artery and Vein.jpg – OpenStax College – CC BY 3.0
Radial pulse.jpg – U.S. Air Force Staff Sgt. Jeanette Copeland – Public domain
Blausen 0256 CoronaryArteries 02.png – BruceBlaus – CC BY 3.0
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